

The Empirical Case for Gaussian Renormalization as Default

Peter Cotton

Provisional draft, 17 August 2026

Abstract

A population's choice shares on a full menu do not tell you its choices from smaller menus without a rule connecting them. Luce's axiom supplies one: renormalize, leaving the odds between survivors untouched. An independent Gaussian race supplies another: re-run the contest without the removed options, which makes the survivors less lopsided. Neither rule needs data from the smaller menus. In an experiment where subjects chose separately from all thirty-one subsets of five goods, the Gaussian rule predicts held-out choices better, and across twelve archival ranking collections it does so in twelve of twelve. Both results survive comparison with a null in which the axiom holds by construction, which matters because contraction of a noisily estimated share vector is also shrinkage. The recommendation is narrow and practical: when mapping aggregate full-menu shares to restricted-menu shares, make the Gaussian race the default.

1 Two rules, one calibration

For a population and a menu S , let p_i be the share choosing or first-ranking item i . Luce's axiom [5] assigns a worth $u(i)$ with

$$p_i = \frac{u(i)}{\sum_{j \in S} u(j)}, \quad (1)$$

so the shares determine the worths up to scale, and the prediction for any subset $T \subseteq S$ follows by changing the denominator alone. Odds between survivors are untouched: for $i, j \in T$ the share preferring i is $p_i/(p_i + p_j)$ when T is the pair. This is Independence of Irrelevant Alternatives.

Thurstone [8] connects the same two quantities through a contest. Item i carries a location a_i , draws a performance $X_i \sim N(a_i, 1)$ on each occasion, and the highest draw is chosen, so

$$p_i = \int \varphi(x - a_i) \prod_{k \neq i} \Phi(x - a_k) dx. \quad (2)$$

The integral is one-dimensional for any number of alternatives, since conditioning on $X_i = x$ makes the other events independent, and it is inverted numerically by a lattice method [2]; locations are identified up to a common translation, fixed here by $\sum_i a_i = 0$. Prediction for a subset T means re-running the contest among T alone. For a pair this is one normal evaluation,

$$P(X_i > X_j) = \Phi\left(\frac{a_i - a_j}{\sqrt{2}}\right), \quad (3)$$

and it is less lopsided than $p_i/(p_i + p_j)$: winning a large field needs an unusually good draw, and a favourite's edge is partly an edge in producing extremes, so that edge counts for less when one rival remains.

Both rules are calibrated to the same $|S| - 1$ free shares and neither sees any observation from a smaller menu. The comparison is therefore between two zero-extra-parameter maps, not between two fitted models. That is worth stating plainly, because it also bounds what the comparison can show. A fitted Plackett–Luce likelihood uses the whole ranking and a fitted Bradley–Terry model uses the pairwise outcomes directly; neither is tested here. What is tested is a cross-moment implication of the homogeneous Luce model, namely whether its full-menu shares correctly determine its restricted-menu shares.

That the direction of the Gaussian disagreement is not an accident can be shown. With the highest draw winning, let $r = f/F$ be the reverse hazard of the common noise. For nested menus $T \subset S$ and $i, j \in T$ with $a_i > a_j$, if $\log r$ is concave then the odds of i against j are at least as large in S as in T , strictly so when $\log r$ is strictly concave and at least one contestant is actually removed. The argument is a covariance inequality between two increasing functions, and the Gaussian satisfies the hypothesis because $\frac{d^2}{dx^2} \log r(x) = -\text{Var}(Z \mid Z < x) < 0$. The Gumbel has $r(x) = s^{-1}e^{-x/s}$, so $\log r$ is affine, the inequality is an equality, and renormalization is recovered exactly; this is the sufficiency half of Yellott's theorem [9], which is what makes the axiom a boundary of the Gaussian's class rather than a rival to it. We use the proposition only for the sign. It gives no ordering of magnitudes, and contraction is not a property of a noise law at all: it depends on the share configuration, and laws recalibrated to the same shares reorder between configurations, so we make no claim of the form heavier tails contract less.

2 What is observed, and at what level

Twelve of the thirteen collections used here are complete rankings. A ranking yields both the first-placed item and the relative order of any two items, so in those twelve the comparison is between two accounts of one ranking distribution rather than a record of behaviour under restriction. A respondent's ranking imposes one ordering on every subset by construction, so no ranking dataset can show a person choosing i over j from one menu and j over i from another, and none reveals whether that person would choose as their ranking implies when handed two options. Any statement about restriction from those twelve rests on the assumption that they would.

The thirteenth makes no such assumption, and for that reason it comes first below.

The level of analysis matters as much as the elicitation. Every quantity here is a population quantity, and aggregate failure of the axiom does not imply that individuals violate it. A population of people who each choose by Equation (1) with worths differing between people generally does not satisfy the axiom in aggregate, which is why mixed logit is useful and why its probabilities are integrals over logit probabilities rather than logit probabilities [7]. So what can be rejected here is a representative-agent Luce restriction map. Individual-level noise is not identified.

3 Choices from real menus

Costa-Gomes, Cueva, Gerasimou and Tejiscak [1] had each subject choose separately from all thirty-one subsets of five consumer goods, so restriction is observed rather than imputed. The

archive names the goods only by two-letter codes and we describe them no more precisely. This is the cleanest available test of the question above, and it is the primary result.

Subjects are split into five folds. On the training subjects, shares come only from the full five-item menu; both rules are calibrated to those shares and nothing else; each then predicts every menu of size two, three and four; and we score the choices actually made by held-out subjects. Intervals resample subjects and repeat the whole procedure, calibration included. Some subjects may defer, so the forced-choice subgroup is reported separately: those subjects cannot defer, and no coding decision about deferral enters that row.

Subjects	n	Choices	Renorm.	Race	Gain
All	364	8,239	0.9263	0.9164	+0.0100 [+0.0048, +0.0246]
Forced choice only	137	3,425	0.9112	0.8846	+0.0265 [+0.0116, +0.0700]

Table 1: Held-out log loss per choice on menus of size two to four, with both rules calibrated only to training-fold full-menu shares under a common add- α convention, $\alpha = 1/2$. Shares rest on 268 full-menu choices for all subjects and 109 for the forced-choice subgroup. Intervals are subject bootstraps that recompute folds, shares, calibration and predictions.

The Gaussian rule is better on both, and better by more where deferral cannot interfere. Section 5 shows the gain is not an artefact of estimating shares from a few hundred observations.

4 Choices induced from rankings

Twelve archival collections have human respondents and rankings, or ratings fine enough to yield them. Two rank-order card tasks from the General Social Survey ask what a child should learn and what matters in a job. Three independent samples of Jester users rate the same ten jokes. Sushi rankings come from a Japanese web survey, film rankings are induced from Netflix ratings, and dots and puzzles are perceptual discrimination tasks. The rest are rankings of seven sports by participation preference, ten occupations by perceived prestige, and four political goals from a five-nation survey. The thirteen collections comprise 147,719 responses, of which the menu experiment contributes 373 subjects; the twelve ranking collections contribute 147,346. These are not thirteen independent experiments, since three are Jester files, two are General Social Survey items, and dots and puzzles are companion tasks.

The protocol matches Section 3 with respondents in place of subjects, and the outcome for a subset is the respondent’s highest-ranked member of it.

Averaging over every subset weights cardinalities by how many there are, so intermediate menus dominate: for $K = 10$ only 45 of 1,013 subsets are pairs, about four per cent of the weight. Those middling menus are where least has been removed and least is at stake. Splitting by menu size shows the pairwise gain exceeding the aggregate in eleven of twelve datasets, by factors from 0.7 to 3.7 with median 1.8: Sushi is +0.0412 at $|T| = 2$ against +0.0111 aggregate, and GSS socialization +0.0130 against +0.0055. Occupational prestige is the exception, peaking at $|T| = 4$. At $|T| = K$ nothing has been removed, the two rules must agree, and the measured gain is zero to four decimals in every dataset, which is the check that the pipeline is sound. We keep the all-subsets aggregate as the headline because it fixes the estimand in advance rather than selecting a cardinality after the fact.

Dataset	n	K	Subsets	Renorm.	Race	Gain
Occupational prestige	143	10	1,013	1.0173	0.9783	+0.0390 [+0.0227, +0.0555]
Sushi	5,000	10	1,013	1.3724	1.3613	+0.0111 [+0.0094, +0.0128]
Political goals	2,262	4	11	0.8835	0.8766	+0.0069 [+0.0059, +0.0078]
GSS socialization	5,000	5	26	0.7984	0.7929	+0.0055 [+0.0041, +0.0070]
Jester file 3	5,000	10	1,013	1.5189	1.5165	+0.0024 [+0.0015, +0.0032]
Jester file 1	5,000	10	1,013	1.5287	1.5267	+0.0020 [+0.0012, +0.0027]
Jester file 2	5,000	10	1,013	1.5247	1.5230	+0.0018 [+0.0010, +0.0025]
GSS job values	5,000	5	26	0.8593	0.8576	+0.0017 [+0.0005, +0.0030]
Dots	3,183	4	11	0.8285	0.8270	+0.0015 [+0.0005, +0.0025]
Netflix	4,379	3	4	0.7932	0.7922	+0.0009 [+0.0007, +0.0012]
Puzzles	3,180	4	11	0.8180	0.8173	+0.0007 [−0.0003, +0.0017]
Sports participation	130	7	120	1.2579	1.2528	+0.0050 [+0.0017, +0.0087]

Table 2: Held-out log loss per prediction over every subset of size two or more. Datasets are capped at five thousand respondents for runtime. The race is lower in twelve of twelve and the interval excludes zero in eleven. Occupational prestige is scorable here only because the shared smoothing removes a zero training cell. Sports participation sits below the rule: its gain is real but Section 5 shows it carries no evidence about the restriction map. Intervals resample respondent losses with the fitted training models held fixed, so unlike Table 1 they omit calibration uncertainty and are too narrow, most seriously in the two smallest rows.

5 Is the gain structural?

A positive gain has two possible sources and log loss cannot tell them apart. The race may be closer to how the population chooses. Or the race may be wrong about that and win anyway, because contracting a noisily estimated share vector is shrinkage, and shrinkage lowers log loss when the input is over-confident. The second mechanism needs no departure from the axiom, so properness of the score licenses nothing here: a proper score says the true probability vector is optimal, not that a plug-in estimate of it beats a biased transformation of the same estimate.

The remedy is to build a population where the axiom holds by construction and run the identical procedure on it. Taking the observed full-menu shares as worths, choices or complete rankings are generated from that exact process, and everything is repeated. Rankings are drawn by sorting $\log u_i + G_i$ for independent standard Gumbel G_i , which is the Plackett–Luce ranking law; for $K \geq 4$ this is a particular joint law consistent with those subset marginals rather than the only one, so the null is specifically a Plackett–Luce ranking null.

The shrinkage mechanism is real and it accounts for the entire gain in the smallest dataset. Sports participation, with 130 respondents, sits exactly on its null and is excluded from every claim here. Everywhere else the mechanism runs against the race, and the excess over the null exceeds the raw gain in all twelve remaining rows. It also rescues puzzles, whose raw interval in Table 2 covers zero while its excess over the null does not. The menu experiment clears the null on both readings, which is the result the paper rests on, since it is the only one where restriction is observed.

Dataset	Observed	Null median	Excess	p
Menu experiment, forced choice	+0.0265	−0.0041	+0.0306	0.010
Menu experiment, all subjects	+0.0100	−0.0023	+0.0123	0.015
Occupational prestige	+0.0390	−0.0227	+0.0617	≤ 0.005
Sushi	+0.0111	−0.0062	+0.0174	≤ 0.005
GSS socialization	+0.0055	−0.0066	+0.0121	≤ 0.005
Political goals	+0.0069	−0.0012	+0.0081	≤ 0.005
GSS job values	+0.0017	−0.0047	+0.0065	≤ 0.005
Jester file 3	+0.0024	−0.0015	+0.0039	≤ 0.005
Jester file 1	+0.0020	−0.0014	+0.0034	≤ 0.005
Jester file 2	+0.0018	−0.0014	+0.0032	≤ 0.005
Dots	+0.0015	−0.0014	+0.0029	≤ 0.005
Puzzles	+0.0007	−0.0011	+0.0019	≤ 0.005
Netflix	+0.0009	−0.0001	+0.0010	≤ 0.005
Sports participation	+0.0050	+0.0051	−0.0000	0.51

Table 3: The same gain when the axiom holds by construction. Two hundred replicates per row, so p is floored at $1/201$; the menu experiment uses its own 373 subjects and menus. Wherever shares rest on more than about two thousand observations the null is negative, because contraction is then a bias rather than useful insurance, so comparing an observed gain with zero understates it.

6 How much of the available improvement is that?

An absolute gain of a few thousandths of a nat is a valid predictive effect size but a poor guide to how much was on offer, because both rules receive the same shares and the shares carry nearly all the predictive content of a restricted menu. To scale it we also score a smoothed subset-frequency forecast, the training-fold frequency with which each item is top within each subset, which is $|T| - 1$ free probabilities per subset.

This benchmark is not an oracle ceiling, and treating it as one would overstate the ratio. It is itself estimated, so its expected held-out loss exceeds the oracle conditional entropy by roughly $(m - 1)/2n$ for an m -outcome menu and n training observations, and a regularized or structured forecast can beat it out of sample. Applying that first-order correction, the Gaussian race closes a median 35 per cent of the empirical benchmark gap, from 16 per cent for sports participation to 58 per cent for dots. Uncorrected the median is 39 per cent. Either way the reading is that a rule with nothing fitted to restricted menus recovers a third or more of what a per-subset frequency estimator achieves, while renormalization recovers none of it, being the unique odds-preserving map. Denominators as small as 0.002 make these ratios unstable, and they are offered as context for the absolute gains rather than as a headline.

7 What this does and does not establish

The practical recommendation is narrow. When the task is to infer restricted-menu population shares from full-menu population shares, the Gaussian race should be the default in place of renormalization. It costs one extra one-dimensional integral, needs no additional data, and was better on every collection examined here except the one too small to discriminate. In the single experiment where restriction is actually observed rather than induced from rankings, it is better

by +0.0265 nats per choice among subjects who cannot defer.

Three things do not follow. Nothing here compares fitted likelihoods: a fitted Plackett–Luce model uses the whole ranking and a fitted Bradley–Terry model uses pairwise outcomes directly, and neither was estimated, so no recommendation about ranking or paired-comparison likelihoods is supported. The held-out score is a sum of marginal scores for top-item-within-subset outcomes, proper for those marginals but not for the full ranking distribution; with $K = 4$ the scored marginals carry at most 17 degrees of freedom against 23 for a distribution over rankings, so distinct ranking laws can agree on everything scored. And nothing identifies individual-level noise, for the reason given in Section 2: heterogeneous Luce choosers reproduce aggregate contraction, so the target rejected is a representative-agent map.

Nesting is also not the same as costlessness. The independent-noise contest class contains the Gumbel point, so permitting the class excludes no Luce population. But Table 3 shows that choosing the Gaussian point specifically is a statistical decision that loses when the axiom is true and the shares are well estimated. The class costs nothing in generality; the default within it is a bet, and the evidence here is that it is a good one for human population data.

What would settle more: scoring the exact Gaussian ordering law against Plackett–Luce ranking probabilities. We do not do that. Pricing ordered outcomes by removing the winner and re-running is a sequential rule, not the race’s ordering law, and against that law it misprices individual pairs and triples by up to a factor of 3.4, so it cannot stand in.

Data and code

Sushi rankings come from Kamishima [4]; Jester ratings from the Eigentaste project [3]; the General Social Survey cumulative file from NORC; the perceptual, leisure, prestige, film and political-goal rankings via PrefLib [6] and the CRAN packages PLMIX and ConsRank; the menu-choice data from the replication archive of [1]. All are public. Analysis code and every input needed to reproduce the tables are under `research/human` in github.com/microprediction/winning, the inputs as column extracts rather than raw archives.

Two storage conventions must be told apart when reproducing this. BayesMallows, ConsRank and PerMallows store ranks, so element j holds the rank of alternative j , while PLMIX stores orderings, so the first element names the favourite. Reading one as the other transposes the data silently and raises no error. We established which is which by cross-validation: the political goals data ships in two of these packages and they agree only when each is read under its own convention.

References

- [1] Miguel A. Costa-Gomes, Carlos Cueva, Georgios Gerasimou, and Matúš Tejvšćák. Choice, deferral and consistency. *Quantitative Economics*, 13:1297–1318, 2022.
- [2] Peter Cotton. Inferring relative ability from winning probability in multientrant contests. *SIAM Journal on Financial Mathematics*, 12(1), 2021.
- [3] Ken Goldberg, Theresa Roeder, Dhruv Gupta, and Chris Perkins. Eigentaste: a constant time collaborative filtering algorithm. *Information Retrieval*, 4(2):133–151, 2001.

- [4] Toshihiro Kamishima. Nantonac collaborative filtering: recommendation based on order responses. In *Proceedings of KDD*, 2003.
- [5] R. Duncan Luce. *Individual Choice Behavior: A Theoretical Analysis*. Wiley, 1959.
- [6] Nicholas Mattei and Toby Walsh. PrefLib: a library of preference data. In *Algorithmic Decision Theory*, 2013.
- [7] Daniel McFadden and Kenneth Train. Mixed MNL models for discrete response. *Journal of Applied Econometrics*, 15(5):447–470, 2000.
- [8] Louis L. Thurstone. A law of comparative judgment. *Psychological Review*, 34:273–286, 1927.
- [9] John I. Yellott. The relationship between Luce’s choice axiom, Thurstone’s theory of comparative judgment, and the double exponential distribution. *Journal of Mathematical Psychology*, 15:109–144, 1977.